

DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

Week: 8

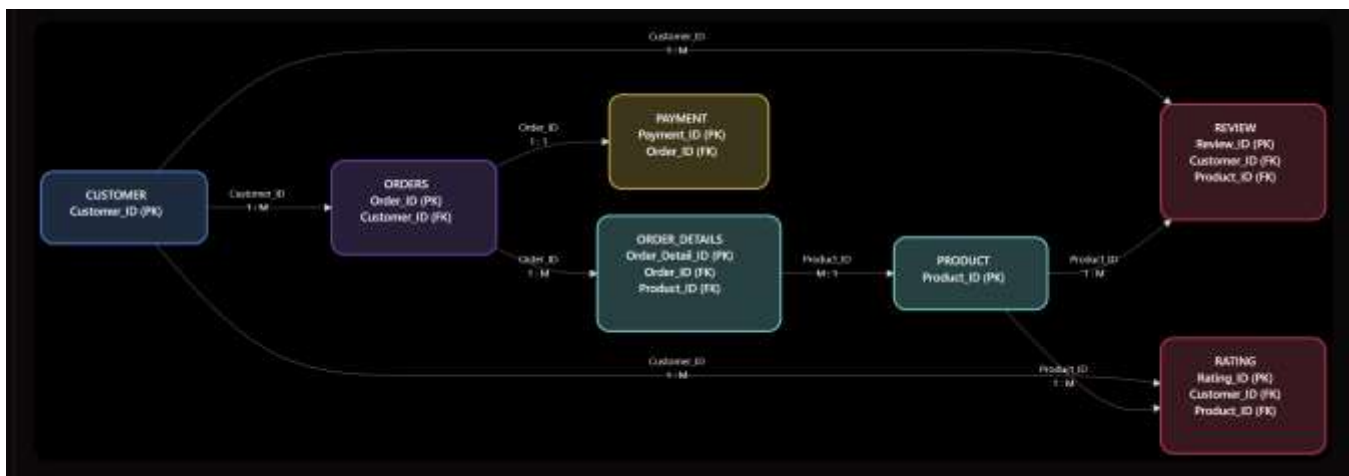
Objectives

1. Combine Customer, Product, Order, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

1. Combining Multiple Tables

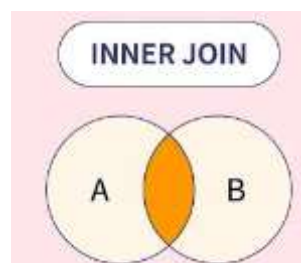
The EcoCart database contains different tables that store customer, product, order, and payment information.

The tables can be combined using **JOIN operations** to retrieve related information.



2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT
  2   c.Customer_ID,
  3   c.Customer_Name,
  4   o.Order_ID,
  5   o.Order_Date,
  6   o.Total_Amount
  7 FROM Customer c
  8 INNER JOIN Orders o
  9   ON c.Customer_ID = o.Customer_ID;
```

CUSTOMER_ID

CUSTOMER_NAME

ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
101		
Anitha Kumar	201 16-SEP-26	2700
102		
Rahul	202 15-SEP-26	1800

CUSTOMER_ID

CUSTOMER_NAME

ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
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This displays customers who have placed orders.

2.2 Orders and Payment

```
SQL> SELECT
  2   o.Order_ID,
  3   o.Order_Date,
  4   o.Total_Amount,
  5   p.Payment_Mode,
  6   p.Payment_Amount,
  7   p.Payment_Status
  8 FROM Orders o
  9 INNER JOIN Payment p
 10   ON o.Order_ID = p.Order_ID;
```

ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_MODE	PAYMENT_AMOUNT
----------	-----------	--------------	--------------	----------------

PAYMENT_STATUS

201	16-SEP-26	2700	UPI	2500
Successful				
202	15-SEP-26	1800	Card	1800
Successful				
203	12-SEP-26	3200	Net Banking	3200
Successful				

ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_MODE	PAYMENT_AMOUNT
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PAYMENT_STATUS

This displays order and corresponding payment details.

2.3 Product and Order_Details

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     od.Order_ID,
  5     od.Quantity,
  6     od.Unit_Price,
  7     od.Subtotal
  8 FROM Product p
  9 INNER JOIN Order_Details od
 10     ON p.Product_ID = od.Product_ID;
```

PRODUCT_ID

PRODUCT_NAME

ORDER_ID	QUANTITY	UNIT_PRICE	SUBTOTAL
101			
Organic Turmeric Powder			
201	3	1000	3000
102			
Black Pepper			
201	1	500	500

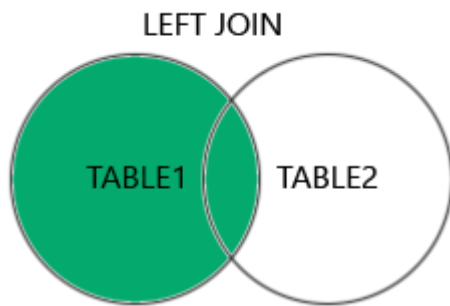
PRODUCT_ID

PRODUCT_NAME

This displays products that have been included in orders.

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders

```
SQL> SELECT
2      c.Customer_ID,
3      c.Customer_Name,
4      o.Order_ID,
5      o.Order_Date,
6      o.Total_Amount
7 FROM Customer c
8 LEFT JOIN Orders o
9      ON c.Customer_ID = o.Customer_ID;
```

CUSTOMER_ID

CUSTOMER_NAME

	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
Anitha Kumar	101		
	201	16-SEP-26	2700

	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
Rahul	102		
	202	15-SEP-26	1800

CUSTOMER_ID

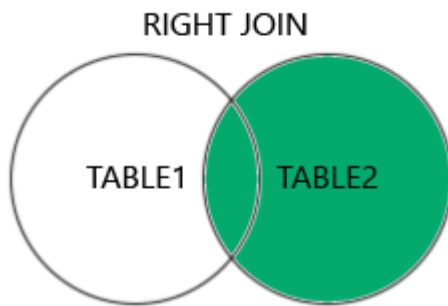
CUSTOMER_NAME

	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT

This displays **all customers**, including customers who have not placed an order.

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

```
SQL> SELECT
2     c.Customer_ID,
3     c.Customer_Name,
4     o.Order_ID,
5     o.Order_Date,
6     o.Total_Amount
7 FROM Customer c
8 RIGHT JOIN Orders o
9     ON c.Customer_ID = o.Customer_ID;
```

CUSTOMER_ID

CUSTOMER_NAME

	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
	101		
Anitha Kumar	201	16-SEP-26	2700
	102		
Rahul	202	15-SEP-26	1800

CUSTOMER_ID

CUSTOMER_NAME

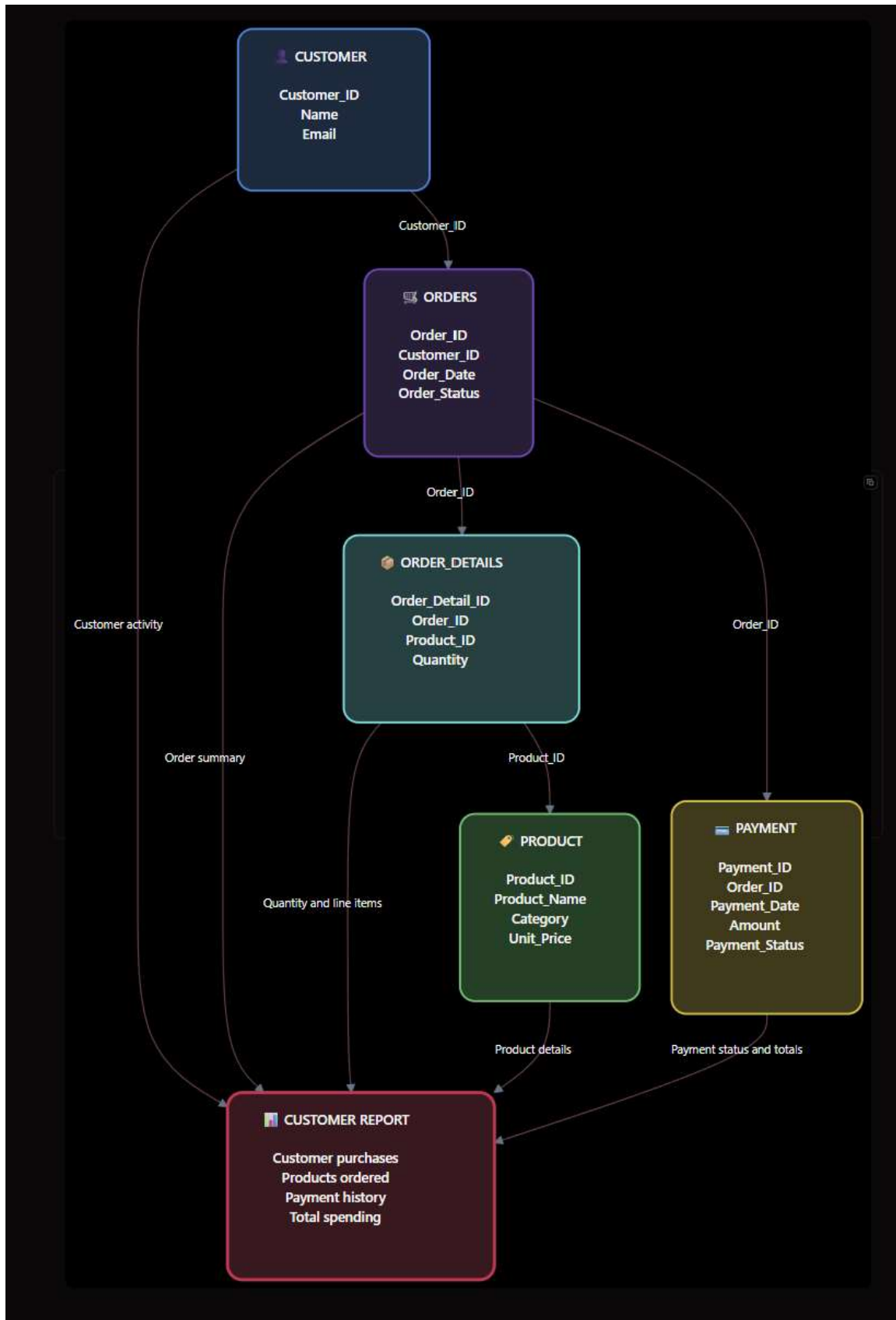
	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
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This displays **all orders** along with the matching customer information.

5. Complete Order Details

This query combines **Customer**, **Orders**, **Order_Details**, **Product**, and **Payment** tables.

```
SQL> SELECT
2      c.Customer_ID,
3      c.Customer_Name,
4      o.Order_ID,
5      o.Order_Date,
6      p.Product_ID,
7      p.Product_Name,
8      od.Quantity,
9      od.Unit_Price,
10     od.Subtotal,
11     py.Payment_Mode,
12     py.Payment_Amount,
13     py.Payment_Status
14 FROM Customer c
15 INNER JOIN Orders o
16     ON c.Customer_ID = o.Customer_ID
17 INNER JOIN Order_Details od
18     ON o.Order_ID = od.Order_ID
19 INNER JOIN Product p
20     ON od.Product_ID = p.Product_ID
21 INNER JOIN Payment py
22     ON o.Order_ID = py.Order_ID
23 ORDER BY o.Order_Date DESC;
```



6. Customer Purchase History

This query displays the products purchased by each customer.

```
SQL> SELECT
  2     c.Customer_ID,
  3     c.Customer_Name,
  4     o.Order_ID,
  5     o.Order_Date,
  6     p.Product_Name,
  7     od.Quantity,
  8     od.Subtotal
  9 FROM Customer c
 10 INNER JOIN Orders o
 11     ON c.Customer_ID = o.Customer_ID
 12 INNER JOIN Order_Details od
 13     ON o.Order_ID = od.Order_ID
 14 INNER JOIN Product p
 15     ON od.Product_ID = p.Product_ID
 16 ORDER BY c.Customer_ID, o.Order_Date;
```

CUSTOMER_ID

CUSTOMER_NAME

ORDER_ID ORDER_DAT

PRODUCT_NAME

QUANTITY SUBTOTAL

101

Anitha Kumar

7. Customer Purchase History with Payment

```
SQL> SELECT
 2      c.Customer_ID,
 3      c.Customer_Name,
 4      o.Order_ID,
 5      o.Order_Date,
 6      p.Product_Name,
 7      od.Quantity,
 8      od.Subtotal,
 9      py.Payment_Mode,
10      py.Payment_Status
11 FROM Customer c
12 INNER JOIN Orders o
13      ON c.Customer_ID = o.Customer_ID
14 INNER JOIN Order_Details od
15      ON o.Order_ID = od.Order_ID
16 INNER JOIN Product p
17      ON od.Product_ID = p.Product_ID
18 INNER JOIN Payment py
19      ON o.Order_ID = py.Order_ID
20 ORDER BY c.Customer_ID;
```

CUSTOMER_ID

CUSTOMER_NAME

ORDER_ID ORDER_DAT

PRODUCT_NAME

QUANTITY SUBTOTAL PAYMENT_MODE

PAYMENT_STATUS

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT
  2     c.Customer_Name,
  3     COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  4     SUM(od.Subtotal) AS Total_Purchase,
  5     SUM(od.Quantity) AS Total_Products
  6 FROM Customer c
  7 INNER JOIN Orders o
  8     ON c.Customer_ID = o.Customer_ID
  9 INNER JOIN Order_Details od
 10     ON o.Order_ID = od.Order_ID
 11 GROUP BY c.Customer_Name
 12 ORDER BY Total_Purchase DESC;
```

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS

Anitha Kumar	1	3500	4
Priya	1	3200	2
Rahul	1	1800	2

9. Payment Transaction Report

```
SQL> SELECT
2      c.Customer_Name,
3      o.Order_ID,
4      o.Order_Date,
5      o.Total_Amount,
6      py.Payment_Mode,
7      py.Payment_Amount,
8      py.Payment_Status
9  FROM Customer c
10 INNER JOIN Orders o
11      ON c.Customer_ID = o.Customer_ID
12 INNER JOIN Payment py
13      ON o.Order_ID = py.Order_ID
14 ORDER BY o.Order_Date DESC;
```

CUSTOMER_NAME

ORDER_ID	ORDER_DAT	TOTAL_AMOUNT	PAYMENT_MODE	PAYMENT_AMOUNT
----------	-----------	--------------	--------------	----------------

PAYMENT_STATUS

Anitha Kumar	201 16-SEP-26	2700	UPI	2500
Successful				
Rahul	202 15-SEP-26	1800	Card	1800
Successful				

CUSTOMER_NAME